

EDGE-AI · RETROFIT INTELLIGENCE

The plug that makes legacy machines AI-smart.

Machines built decades before AI can now sense the fault building, explain it, and act — **before anyone gets hurt, with the cable unplugged.**

Plug-in retrofit

Fully offline AI

Autonomous actuation

THE PROBLEM · TODAY

The world runs on machines built before AI — and they can't say when something's about to kill.

Faults, fires and gas build up unnoticed until the last minute — replacement costs six figures per machine; awareness shouldn't.



America's 5th-deadliest job Waste collection: 36 worker deaths in 2024 (BLS). Equipment and transport incidents lead the causes.



Worst fire year on record 448 waste-facility fires in 2025, ~\$2.5B in losses — lithium batteries and vapes hiding in the load, unseen until they ignite.



Gas that kills the rescuers too 46 US workers died from H₂S (2011–17). At one landfill leachate station, one worker collapsed — three more died trying to save him.

THE GAP

Today's retrofits ship data away to think.

Cameras, modems and fault codes all assume a **cloud and coverage** — but remote routes, pits and fields are exactly where connectivity dies:

- Fault codes fire at/after the breach — not before
- Camera + DTC systems diagnose in the cloud
- Nothing acts locally when the link is down
- The machine itself stays dumb

THE PRODUCT

A retrofit AI module for machines that predate AI

CLAMP-ON SENSOR POD

Sense the machine

- Vibration · temp · acoustic (today)
- Gas pod: H₂S / CH₄ (roadmap)
- or tap the J1939 / CAN bus directly



THE AI MODULE · PI-5 CLASS

Think on the machine

- Detect: 3-tier, deterministic
- Diagnose: on-device LLM
- Decide: pure local policy



ACTUATION & ALERTING

Act — even offline

- Alarm / cutoff relay / servo
- Operator phone alert + dashboard
- served from its own hotspot

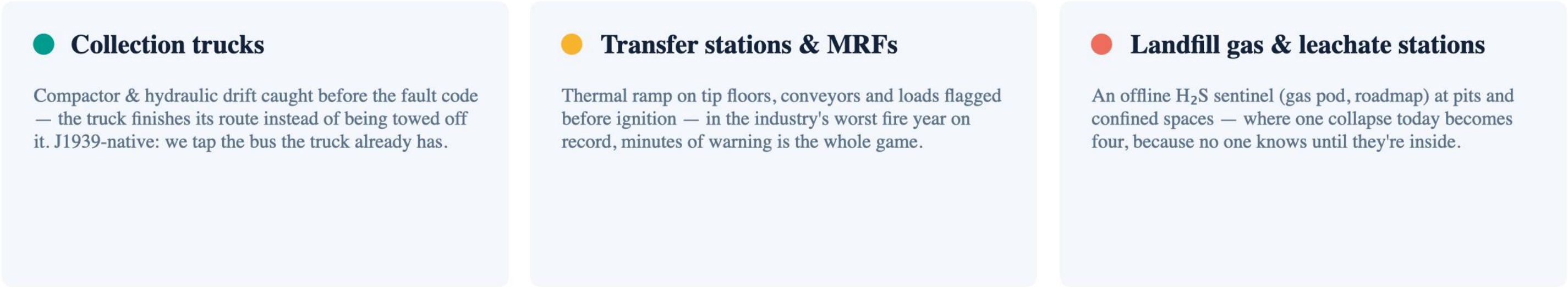
Installs in minutes on any make or age of machine · **~\$110 hardware** · no cloud contract · uplinks **~3 KB of insight**, not a **~121 KB firehose**

Key choice: detection is deterministic and never depends on a network or the LLM — the machine protects itself even when nothing else is reachable.

Republic Services: 17,800 trucks in America's 5th-deadliest job



THREE DEPLOYMENT SURFACES — SAME MODULE



Target-customer analysis from public data (SEC FY2025 filing, BLS, Fire Rover, OSHA/NIOSH) — chosen because every pain maps to what this module senses.

ONE SHIFT WITH THE MODULE

Three moments on a route — and what you'll see live

06:12

Packer hydraulics start to drift

CUSUM flags slow wear long before any code trips — 193 s of lead in our measured run. The truck finishes the route and rolls into a planned bay, not onto a tow hook.

LIVE: the shake you'll see = this drift

09:40

Heat ramp in the hopper

A battery in the load starts cooking. Thermal ramp breaches the bound → ALARM: cab alert, NeoPixel flash + tone at the sensor, phone vibrates — before the tip floor, not after.

LIVE: warming the sensor = this precursor

13:05

Worker check-in at the leachate station

Manual trigger opens an inspection event, honestly labeled — never blended into sensor data. With the gas pod (roadmap), this becomes the H₂S sentinel that stops a rescue-chain tragedy.

LIVE: Button A = this check-in

Same detection core, same dashboard, same hardware you are about to watch — only the machine around it changes.

One offline loop: **perceive** → **decide** → **act**

1

Perceive

The sensor pod streams vibration/temp/acoustic at 10 Hz; on J1939 machines we read the bus directly. (Demo: CPX board over USB serial + reproducible J1939 CSVs.)

2

Detect

Three parallel detectors — fixed threshold, IsolationForest, CUSUM drift. ~4 ms/row, no GPU, no network.

3

Diagnose

A small on-device LLM (qwen2.5:1.5b via Ollama) writes a 2-sentence operator summary — deterministic template fallback.

4

Decide → Act

A pure decide() policy drives the actuator: alarm, relay, servo — plus phone alert. Degrades to a logging mock, never silently.

The right tool for each job — and we can defend every choice

CLASSICAL ML

Detection — no LLM, on purpose

The safety path must be deterministic, fast, and incapable of hallucinating a fault or missing a red-line.

- Fixed threshold — cannot be talked out of a breach
- IsolationForest — multivariate outliers
- CUSUM — slow predictive drift
- No GPU · no network · no prompt

EDGE LLM

Diagnosis — where language is the task

Turning a 4-channel numeric state into a sentence an operator can act on is a translation problem.

- Exactly what a small LM is good at
- Quarantined as the one fail-allowed layer
- Never gates detection or actuation
- Latency & non-determinism can't endanger safety

AI where it adds value · determinism where lives depend on it.

Why qwen2.5:1.5b — chosen by measurement, not reputation

THE ON-DEVICE SWEEP

Each candidate timed on the Pi 5 on the real anomaly→summary prompt (benchmark_edge_llm.py). Deciding metric: time to a 2-sentence summary vs. a 10 s cutoff.

qwen2.5:0.5b	<i>fast, but lower quality</i>	✓
qwen2.5:1.5b	<i>speed / quality sweet spot</i>	✓ CHOSEN
llama3.2:1b	<i>under the cutoff</i>	✓
gemma2:2b	<i>under the cutoff, larger</i>	✓
llama3.2:3b	<i>too slow on the Pi</i>	✗

WHY IT'S THE RIGHT PICK

- **Measured, not assumed**
The benchmark times each model on the exact prompt the demo uses — the choice is data, not a spec-sheet.
- **Runs ON the module (Option A)**
~6–12 s live, under the 10 s cutoff → no gateway, no cloud, no network in the loop.
- **Speed / quality sweet spot**
Big enough for a coherent 2-sentence diagnosis, yet 4-bit-quantized & CPU-only — shares 4 cores with the stream loop + Flask. 3B was too slow; 0.5B weaker.
- **Small is safe here**
A deterministic template fallback covers the rare slow call, so we optimize for usually-fast + coherent, not worst-case.

Why not a bigger model? It can't meet the on-device budget — and the task doesn't need it. We measured both.

ALREADY REAL · MEASURED RESULTS

Every number reproducible on this module today

193 s

PREDICTIVE LEAD

CUSUM flagged pump drift before the hard safety threshold broke

~97 %

LESS UPLINK

~121 KB raw telemetry → ~3 KB AI-summarized insight per 30-min window

13 / 13

FAULT WINDOWS

0 false alarms · 0 soft flags · 90 noise blips debounced on the healthy machine

~4 ms

PER ROW

~250× real-time headroom at 1 Hz — streaming inference, not batch

6–12 s

PER DIAGNOSIS

Median edge-LLM latency for a 2-sentence operator summary

\$0

CLOUD BILL

No contract, no recurring inference cost, no connectivity requirement

Zero alert fatigue — every alert an operator sees is real.

Real hardware, verified — not a slideware demo

● Runs on real silicon

Sensor firmware on CircuitPython 10.2.1; a 447-frame live capture confirms all four signals. We found & fixed a platform bug (cp.sound_level unsupported → drive the PDM mic directly).

● Loop closes at the sensor

On a confirmed fault the module writes back over serial → the sensor pod flashes + sounds at the machine. Fixed a second platform trap: CircuitPython input() froze the sample loop; replaced with a raw byte-buffer read.

● Mobile-first deployment

The module hosts its own WiFi hotspot + local DNS — a phone gets the live dashboard at a plain hostname, no IP, no port, zero external network.

● Graceful degradation everywhere

GPIO → logging mock; LLM → template on timeout; serial skips malformed lines; unconfirmed events get an honest 'monitor' note. No silent failure mode.

● Validated, not hand-tuned

All hysteresis params fixed by a sweep across four scenarios (healthy = 0 events, every fault caught, ≥60 s lead). Fixed seeds regenerate every dataset identically.

“We found and fixed two real CircuitPython platform bugs getting this working — this is a built system, not a mockup.”

DIFFERENTIATION

They retrofit telemetry. We retrofit intelligence.

Deployed retrofit tech	What it does	Where we differ
Heil / 3rd Eye Connected Collections	Cameras + engine DTCs → cloud analytics, tied to Heil bodies	On-device diagnosis · body-agnostic · works offline
McNeilus NGEN body controls	Advanced diagnostics on their own new bodies	Retrofits any make or age — including the old fleet
Deere JDLink M-Modem legacy telematics	Uplinks location & machine data from old tractors to the cloud	Intelligence on the machine — works with zero coverage
OBD / DTC codes on-board diagnostics	A code fires at/after the threshold breach	CUSUM flags the drift before the code
1. Thinks on the machine Generative diagnosis on-device, offline 2. Acts on the machine Alarm / cutoff with zero network 3. Predicts before the code Drift caught before the hard bound		

BUSINESS CASE

The pilot we'd pitch Republic on Monday

THE ECONOMICS

- **~\$110 vs. \$500k–\$10M+** hardware per machine vs. the cost range of one facility fire event (2025 industry: ~\$2.5B total)
- **A tow costs more than a garage of modules** unplanned roadside failures vs. planned bay visits, at tens of \$k per downtime-hour
- **~97% uplink cut** makes cellular-poor rural routes and remote stations viable — no new connectivity spend

PILLAR FIT

Mobility

Sustainability

Tech

Physical AI

Agentic AI

90-DAY PILOT

1 garage · 50 trucks · 1 transfer station

- Faults caught before the DTC ever fires
- False-alarm rate (ours today: 0 across 13/13 windows)
- Avoided tows & unplanned downtime hours
- Thermal precursor alerts on the tip floor

THEN THE SAME MODULE EXPANDS

WM (~2× the trucks) → municipal fleets → **the legacy farm fleet** (autonomy retrofits cost \$50–150k; we're the \$110 awareness layer for decades-old tractors) → rigs & remote energy.

ROADMAP

What ships next on the same module

NEXT · PRODUCT CONFIG

Gas-sensor pod (H₂S · CH₄ · CO)

The confined-space sentinel for leachate stations, pits and transfer floors — **the alert that goes out before the rescuer goes in.**

IN PROGRESS

Remaining Useful Life (RUL)

Reuse the CUSUM drift slope to say *“hydraulic pressure crosses the bound in ~4 h at this wear rate.”* Lead-time becomes plannable.

IN PROGRESS

On-device service-manual RAG

Ground the LLM in the machine's manual so the diagnosis names the **part, procedure, and torque spec** — all offline.

Honesty line: today's demo-verified sensing is vibration · temp · acoustic · manual trigger. The gas pod is a **hardware config on the identical pipeline** — same detectors, same dashboard, same actuation. Everything else you saw is built & verified.

THE PITCH

Old iron. New brain. Republic's 17,800 trucks get AI without buying a single new one — and no one learns about the gas, the fire, or the failure at the last minute.

Predictive

drift caught 193 s early

Diagnostic

plain language, on-device

Autonomous

acts with no network

Retrofit

any make, any age, ~\$110

Speaker Notes & Q&A — v2 (Republic Services)

Presenter reference — talking points, judge-Q&A answers, and data sources per slide.

SLIDE 1 Title — the plug that makes legacy machines AI-smart

v2 framing: we are no longer a generic PdM demo — we're a retrofit AI module that plugs into legacy machines and makes them self-aware. Everything you'll see runs live on this Pi 5. Our flagship target analysis: Republic Services' waste fleet, where the stakes are lives, fires, and gas.

SLIDE 2 The Problem — machines built before AI can't warn anyone

The problem, sharpened for v2: legacy machines can't warn anyone. The waste industry makes it concrete — 5th-deadliest job (BLS 2024, 36 deaths), a record 448 facility fires in 2025 worth ~\$2.5B (Fire Rover annual report), and H₂S that killed 46 workers in 2011–17 including multi-fatality rescue chains (OSHA/NIOSH). Incumbent retrofits are telemetry: they ship data to the cloud to think, and act on nothing locally.

SLIDE 3 The Product — retrofit AI module (sense / think / act)

The product: a clamp-on/plug-in module. Three parts: a sensor pod (vibration, temp, acoustic today — gas pod for H₂S/CH₄ is the near-term roadmap config), the Pi-5-class AI module that detects, diagnoses and decides on-device, and an actuation/alert layer that works with zero connectivity — it even hosts its own hotspot dashboard. ~\$110 of hardware, no cloud contract. On J1939 machines (refuse trucks, tractors) we can tap the bus directly — our pipeline already speaks J1939.

SLIDE 4 Flagship target — Republic Services

Why Republic Services: ~17,800 collection vehicles (FY2025 SEC filing), operating in the 5th-deadliest US occupation (BLS 2024), through the industry's worst fire year on record (448 fires, ~\$2.5B — Fire Rover), with documented H₂S multi-fatality events at exactly the kind of stations we'd instrument. One module, three deployment surfaces: trucks (hydraulic/compactor drift via the J1939 bus they already have), facilities (thermal precursors before tip-floor fires), and gas/leachate stations (offline H₂S sentinel — the roadmap gas pod). Honesty: this is target-customer analysis from public data, not a claimed relationship.

SLIDE 5 One shift with the module — demo mapping

This slide converts the live demo into the customer's day. 06:12 — hydraulic drift = the shake test, with our measured 193 s lead. 09:40 — thermal ramp = the warming test, mapped to the lithium-battery fire story. 13:05 — Button A = the worker check-in, honestly labeled as manual, and pointing to the roadmap gas pod for the H₂S sentinel role. Close the slide with: same core, same dashboard — only the machine changes.

SLIDE 6 How It Works — one offline loop

The loop runs entirely on the module: perceive (sensor pod or the machine's own J1939 bus), detect with classical ML in ~4 ms, diagnose in plain language with the on-device LLM, then decide and

physically act. Detection never waits on the LLM or the network.

SLIDE 7 AI Appropriateness — right tool for each job

We deliberately don't use an LLM for detection — that path must be deterministic and fast. The LLM only does what language models are actually good at: translating numbers into an operator-ready sentence. And it's quarantined so it can never endanger the safety loop.

Q&A — 'What if the LLM gives a wrong answer?' Two cases. (1) It fails loudly (error/timeout): we drop to a deterministic templated summary built from the detected facts — blunter but correct. (2) It hallucinates a plausible-but-wrong diagnosis: that's an advisory-text error, not a safety event. `decide()` acts on the CONFIRMED sensor event, not on the model's words, so a wrong sentence can't miss a fault, can't trigger or suppress the actuator, and can't corrupt the record — the real signal, values and lead time are shown right next to the prose to contradict it. We also constrain the prompt to name the flagged signal, and unconfirmed events get an honest 'monitor' note. Worst case = one misleading sentence beside the real numbers. If pushed: yes it could briefly mislead an operator — which is why on-device service-manual RAG is on the roadmap.

SLIDE 8 Model Choice — why qwen2.5:1.5b

Backup slide for the model-choice question. We didn't pick the biggest model that fit — we benchmarked candidates on-device on the real prompt and picked the one that meets the latency budget with enough quality. qwen2.5:1.5b runs on the Pi in ~6–12 s (llama3.2:3b misses the 10 s cutoff), is quantized and CPU-only so it shares cores with the rest of the pipeline, and the deterministic fallback lets us stay small because it absorbs the rare slow call. If asked for exact per-model numbers: `benchmark_edge_llm.py` prints them — we can run it live.

SLIDE 9 Measured Results — already real

These aren't projections — every number is measured on this hardware and reproducible. 193 s of predictive lead on slow wear is the difference between a planned bay visit and a tow. 97% less uplink is what makes cellular-poor routes viable. And zero false alarms is what keeps drivers from muting the thing.

SLIDE 10 Engineering & Build Quality

This is a working build, not slideware. Real firmware, a 447-frame live capture, two genuine platform bugs found and fixed, mobile deployment over the module's own hotspot, and graceful degradation at every layer so it never fails silently. Lead with the bugs — you don't hit platform bugs in a mockup.

SLIDE 11 Differentiation — telemetry vs. intelligence

We researched the actual incumbents in this vertical. Heil/3rd Eye Connected Collections and McNeilus NGEN are real and good — but they're camera+DTC systems that diagnose in the cloud and are tied to their own truck bodies. Deere's JDLINK M-Modem proves demand for legacy retrofit — but it's a telematics uplink, not intelligence. Nobody thinks AND acts on the machine itself, offline, on any make or age. That's our lane.

SLIDE 12 Business Case — the 90-day Republic pilot

The ask we'd make of Republic: a 90-day pilot — one garage, 50 trucks, one transfer station. KPIs that map to money and safety: faults caught before the DTC, false-alarm rate (ours is zero today),

avoided tows, thermal precursor alerts. Economics: ~\$110 of hardware per machine against fire events that ran the industry ~\$2.5B in 2025 and tows that cost more per hour than a whole garage of modules. Then the same unchanged module expands: WM, municipal fleets, the legacy farm fleet — where autonomy retrofits cost \$50–150k and we're the \$110 awareness layer — then rigs and remote energy.

SLIDE 13 Roadmap — gas pod, RUL, manual RAG

Roadmap, honestly separated: the gas pod ($\text{H}_2\text{S}/\text{CH}_4/\text{CO}$) is the near-term product config — the confined-space sentinel that sends the alert before the rescuer goes in. It's a sensor swap on the identical pipeline, not new intelligence. RUL turns our drift slope into hours-to-failure. Manual RAG makes the diagnosis name the exact part and torque spec, offline. And say the honesty line out loud: today's verified sensing is vibration/temp/acoustic/manual — judges reward knowing exactly what's built vs. next.

SLIDE 14 Close — Old iron. New brain.

Close on the reframe: old iron, new brain. Republic's entire fleet gets AI without buying a single new truck — and nobody finds out about the gas, the fire, or the failure at the last minute. Predictive, diagnostic, autonomous, retrofit. Thank you.

SOURCESfor the market/safety numbers

Republic Services fleet & facility data: FY2025 annual report (SEC EDGAR). Fatality ranking: BLS 2024 census via Waste Dive. Facility fires: Fire Rover annual report 2025 (448 fires, ~\$2.5B) via Resource Recycling. H_2S deaths: BLS 2011–17 via OSHA; multi-fatality leachate-station case: NIOSH FACE reports. Ag retrofit kit pricing (\$50–150k) and adoption: Mordor Intelligence, USDA ERS. Deere JDLink M-Modem: deere.com. Heil/3rd Eye Connected Collections: heil.com, 3rdeyecam.com.